Week 3 Assignment Report

🎯 **Objective**

The objective of this assignment is to develop practical skills in SQL and Microsoft Excel for Data Analytics. Students will learn how to retrieve, filter, sort, group, and analyze data using SQL, while also learning essential Excel functions and tools for data analysis and reporting.

Task 1: Introduction to Databases & SELECT Statement

Using a sample database/table: Understand the basic concept of databases. Create or use a sample table such as Students, Employees, or Sales. Write SQL queries to display all records. Select specific columns using SELECT. Use column aliases where appropriate.

-------------------------------------explanation---

A database is an organized collection of data that can be stored, managed, and retrieved efficiently. Data in a database is commonly stored in tables consisting of rows and columns.

The SELECT statement is used to retrieve data from a table. It can be used to display all records or only specific columns. Column aliases are used to give temporary, meaningful names to columns in the query result.

-----------------------------------code---

```
USE employee;

CREATE TABLE employees (

   employee_id INT PRIMARY KEY,

   name VARCHAR(50),

   age INT,

   department VARCHAR(50),

   salary INT

);

INSERT INTO employees (employee_id, name, age, department, salary)
```

VALUES

(1, 'Aarav', 25, 'IT', 40000),

(2, 'Riya', 24, 'HR', 35000),

(3, 'Rahul', 27, 'Finance', 45000),

(4, 'Priya', 26, 'IT', 50000),

(5, 'Karan', 28, 'Marketing', 42000);

SELECT * FROM employees;

SELECT name, department

FROM employees;

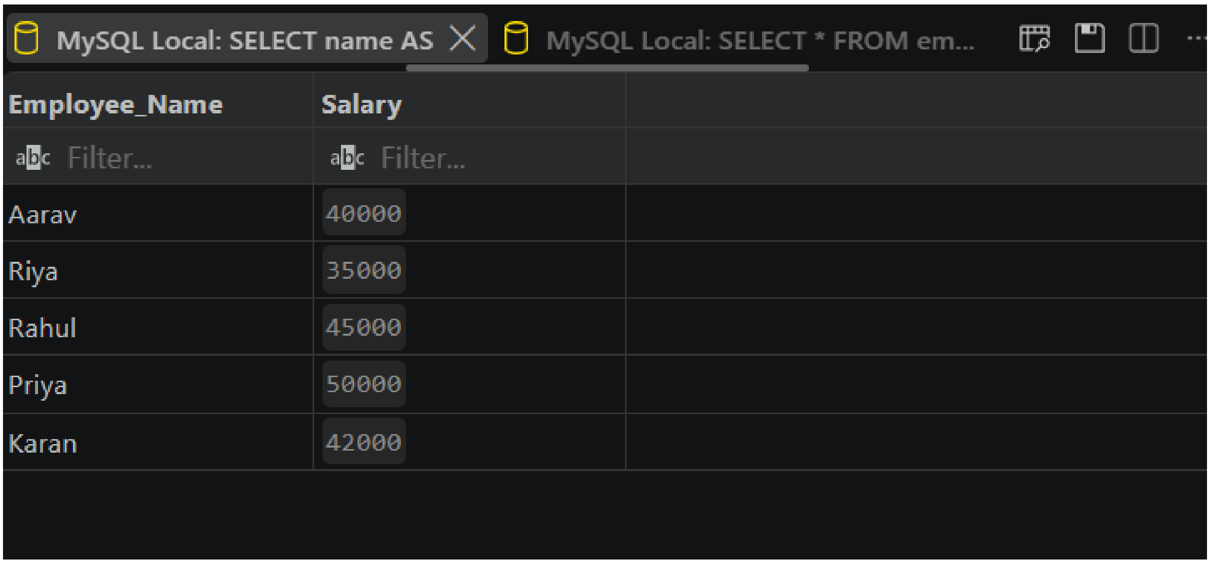
SELECT name AS Employee_Name,

salary AS Salary

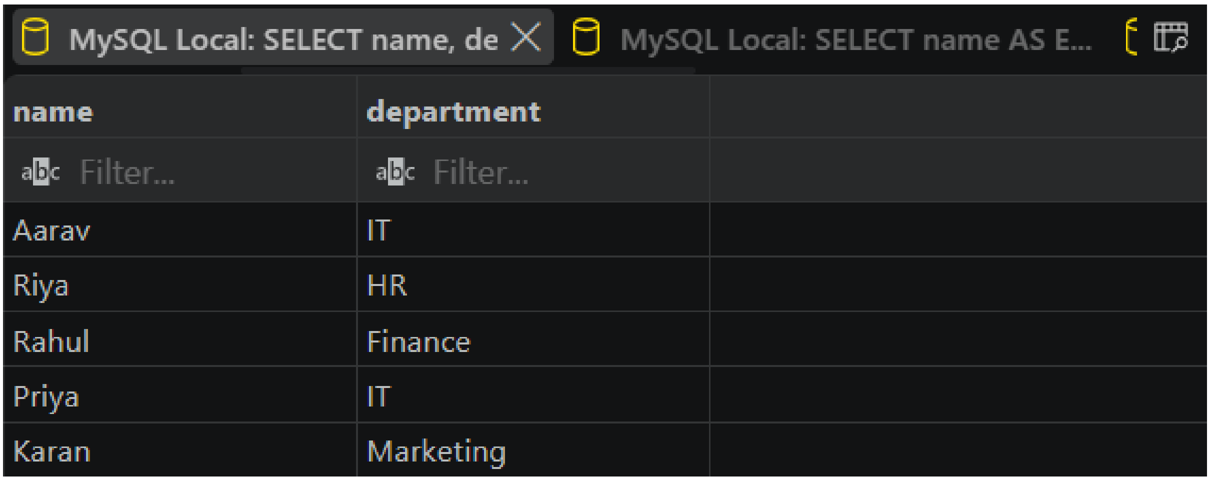
FROM employees;

---**output**---

employee_id	name	age	department
Filter...	Filter...	Filter...	Filter...
1	Aarav	25	IT
2	Riya	24	HR
3	Rahul	27	Finance
4	Priya	26	IT
5	Karan	28	Marketing



Employee_Name	Salary
Aarav	40000
Riya	35000
Rahul	45000
Priya	50000
Karan	42000



name	department
Aarav	IT
Riya	HR
Rahul	Finance
Priya	IT
Karan	Marketing

Task 2: WHERE, ORDER BY & Aggregate Functions

Write SQL queries to: Filter records using the WHERE clause. Use comparison operators in conditions. Sort records using ORDER BY. Perform calculations using aggregate functions: COUNT() SUM() AVG() MIN() MAX() Display the query and its output.

-----------------------------------code--

```
USE employee;

SELECT *

FROM employees

WHERE department = 'IT';

SELECT *

FROM employees

WHERE salary > 40000;
```

```sql
SELECT *
FROM employees
WHERE age >= 26;
SELECT *
FROM employees
ORDER BY salary ASC;
SELECT *
FROM employees
ORDER BY salary DESC;
SELECT COUNT(*) AS Total_Employees
FROM employees;
SELECT SUM(salary) AS Total_Salary
FROM employees;
SELECT AVG(salary) AS Average_Salary
FROM employees;
SELECT MIN(salary) AS Minimum_Salary
FROM employees;
SELECT MAX(salary) AS Maximum_Salary
FROM employees;
```

---**output**---

MySQL Local: multiple query results | MySQL Local: SELECT * FROM e...

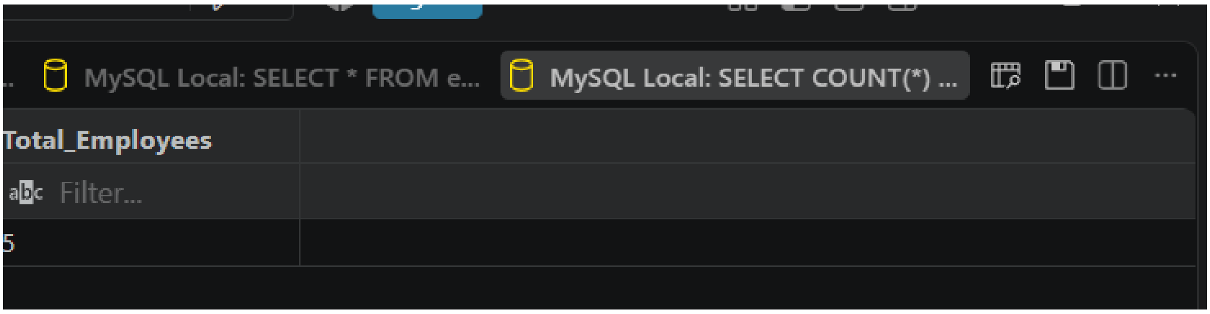
employee_id	name	age	department
Filter...	Filter...	Filter...	Filter...
1	Aarav	25	IT
4	Priya	26	IT

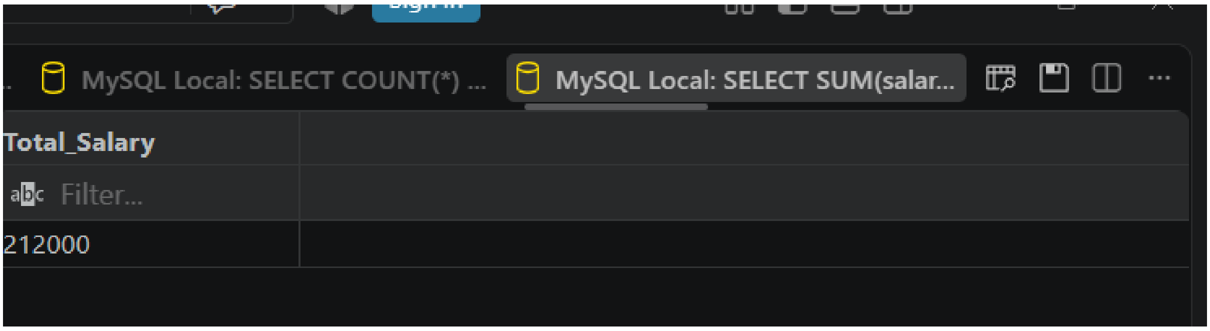
employee_id	name	age	department
Filter...	Filter...	Filter...	Filter...
3	Rahul	27	Finance
4	Priya	26	IT
5	Karan	28	Marketing

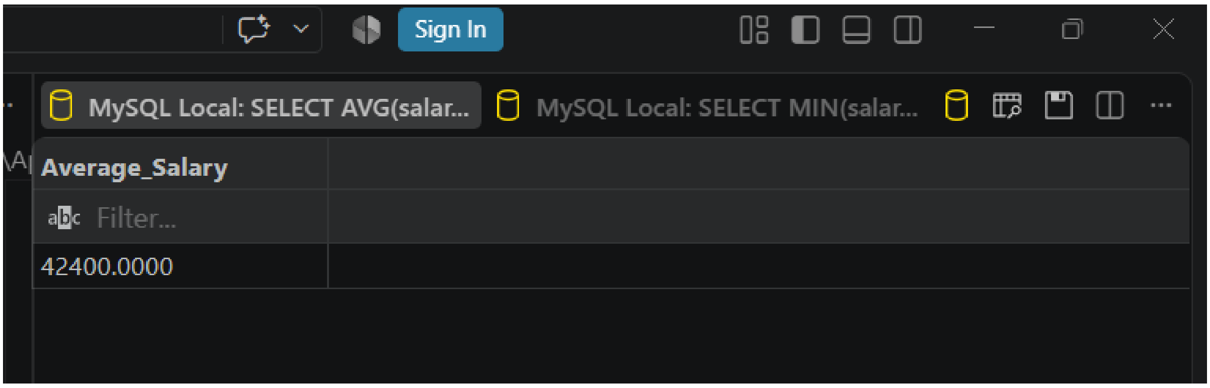
employee_id	name	age	department
Filter...	Filter...	Filter...	Filter...
3	Rahul	27	Finance
4	Priya	26	IT
5	Karan	28	Marketing

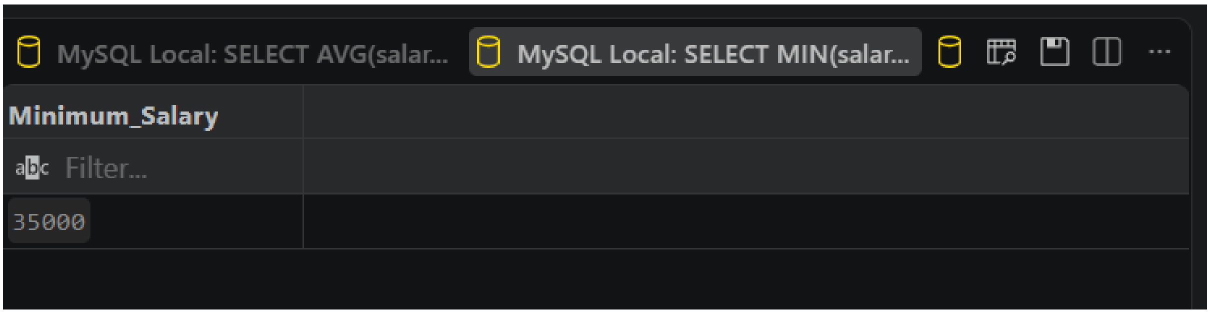
employee_id	name	age	department
Filter...	Filter...	Filter...	Filter...
2	Riya	24	HR
1	Aarav	25	IT
5	Karan	28	Marketing
3	Rahul	27	Finance
4	Priya	26	IT

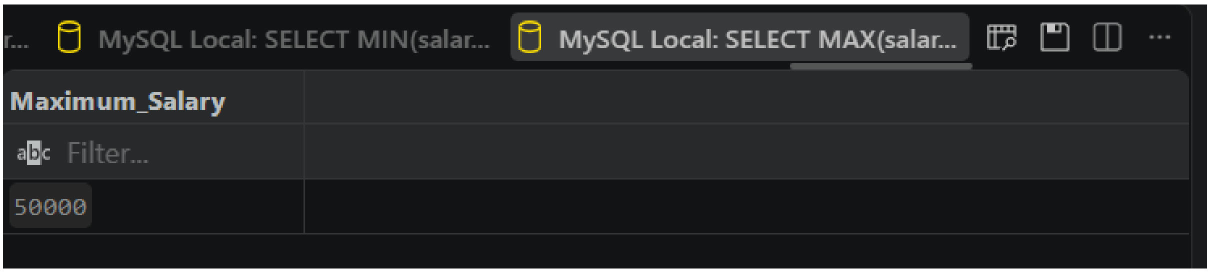
employee_id	name	age	department
Filter...	Filter...	Filter...	Filter...
4	Priya	26	IT
3	Rahul	27	Finance
5	Karan	28	Marketing
1	Aarav	25	IT
2	Riya	24	HR


MySQL Local: SELECT * FROM e...
MySQL Local: SELECT COUNT(*) ...
Total_Employees
Filter...
5


MySQL Local: SELECT COUNT(*) ...
MySQL Local: SELECT SUM(salar...
Total_Salary
Filter...
212000


Sign In
MySQL Local: SELECT AVG(salar...
MySQL Local: SELECT MIN(salar...
Average_Salary
Filter...
42400.0000


MySQL Local: SELECT AVG(salar...
MySQL Local: SELECT MIN(salar...
Minimum_Salary
Filter...
35000


MySQL Local: SELECT MIN(salar...
MySQL Local: SELECT MAX(salar...
Maximum_Salary
Filter...
50000

Task 3: GROUP BY & HAVING

Using a suitable dataset: Group records using GROUP BY. Calculate aggregate values for each group. Use the HAVING clause to filter grouped results. Display and explain the results.

-----------------------------------explanation--

GROUP BY is used to group records that have the same value in a particular column. Aggregate functions such as COUNT() and AVG() can then be used to calculate values for each group.

HAVING is used to filter the grouped results. In this task, it filters departments based on the number of employees in each department.

------------------------------------code---

```
USE employee;

DESC employees;

SELECT department, COUNT(*) AS total_employees

FROM employees

GROUP BY department;

SELECT department, AVG(salary) AS average_salary

FROM employees

GROUP BY department;

SELECT department, COUNT(*) AS total_employees

FROM employees

GROUP BY department

HAVING COUNT(*) > 1;

SELECT department, AVG(salary) AS average_salary

FROM employees

GROUP BY department

HAVING AVG(salary) > 40000;
```

---output---

epartme × MySQL Locals: SELECT departmen... MySQL Locals: SELECT departmen...

department	total_employees
IT	2

MySQL Locals: SELECT departmen... MySQL Locals: SELECT departmen...

lepartment	average_salary
inance	45000.0000
T	45000.0000
larketing	42000.0000

MySQL Locals: SELECT departmen...

department	average_salary
Finance	45000.0000
HR	35000.0000
IT	45000.0000
Marketing	42000.0000

department	total_employees
IT	2

department	average_salary
Finance	45000.0000
IT	45000.0000
Marketing	42000.0000

Task 4: SQL Joins

Create or use at least two related tables and perform: INNER JOIN LEFT JOIN RIGHT JOIN Use a common column to join the tables and display the results. Briefly explain the purpose of each JOIN used.

-----------------------------------explanation--

INNER JOIN: Returns only the records that have matching values in both tables.

LEFT JOIN: Returns all records from the left table and the matching records from the right table. If there is no match, NULL is displayed.

RIGHT JOIN: Returns all records

-----------------------------------code---

```
USE employee;

CREATE TABLE departments (

   department VARCHAR(50),

   location VARCHAR(50)

);

INSERT INTO departments (department, location)

VALUES

('IT', 'Ahmedabad'),

('HR', 'Mumbai'),

('Finance', 'Delhi'),

('Marketing', 'Pune');

SELECT * FROM departments;

SELECT e.employee_id, e.name, e.department, e.salary, d.location

FROM employees e

INNER JOIN departments d

ON e.department = d.department;

SELECT e.employee_id, e.name, e.department, e.salary, d.location

FROM employees e
```

```
LEFT JOIN departments d

ON e.department = d.department;

SELECT e.employee_id, e.name, e.department, e.salary, d.location

FROM employees e

RIGHT JOIN departments d

ON e.department = d.department;
```

--output---

department	location
Filter...	Filter...
IT	Ahmedabad
HR	Mumbai
Finance	Delhi
Marketing	Pune

SELECT * FROM departments; SELECT e.employee_id, e.name, e.departm...

department	location
Filter...	Filter...
IT	Ahmedabad
HR	Mumbai
Finance	Delhi
Marketing	Pune

MySQL Locala: multiple query results | MySQL Locala: SELECT e.employe...

mployee_id	name	department	salary
1	Aarav	IT	40000
4	Priya	IT	50000
2	Riya	HR	35000
3	Rahul	Finance	45000
5	Karan	Marketing	42000

Sign in | Update

MySQL Locala: SELECT e.employe... | MySQL Locala: SELECT e.employe...

employee_id	name	department	salary
1	Aarav	IT	40000
2	Riya	HR	35000
3	Rahul	Finance	45000
4	Priya	IT	50000
5	Karan	Marketing	42000

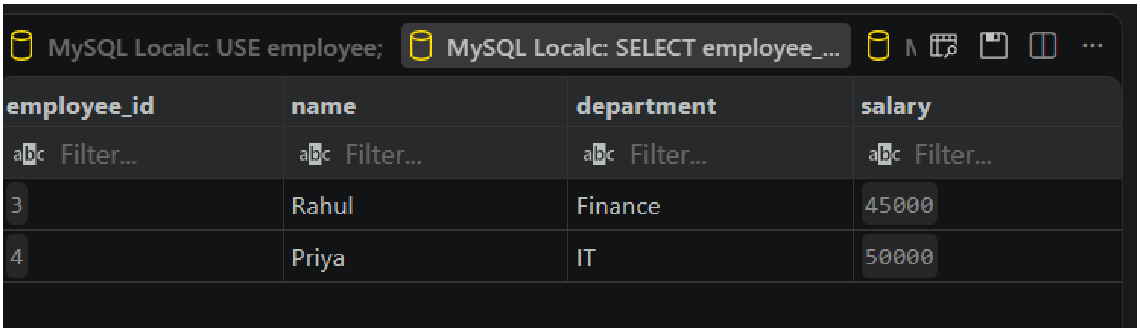
Task 5: SQL Subqueries

Write SQL queries using subqueries to solve at least two problems. Example: Find employees earning more than the average salary. Find products with a price higher than the average product price. Display the query and output. Excel Tasks

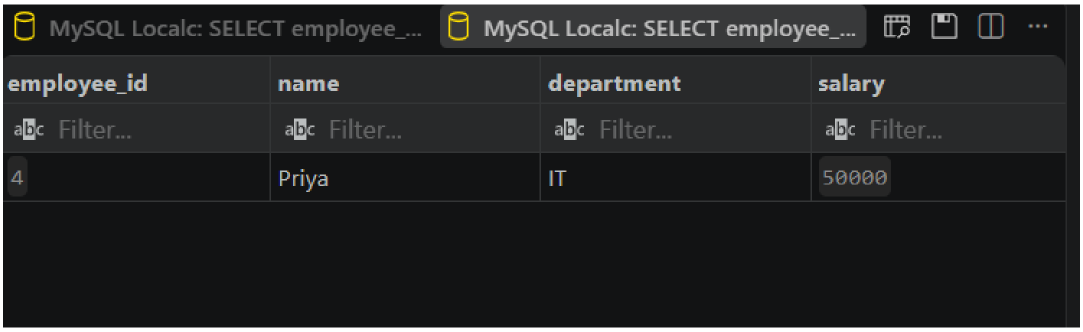
-----------------------------------code--

```
USE employee;

SELECT employee_id, name, department, salary

FROM employees

WHERE salary > (

   SELECT AVG(salary)

   FROM employees

);

SELECT employee_id, name, department, salary

FROM employees

WHERE salary = (

   SELECT MAX(salary)

   FROM employees);
```

---------------------------------------output---



employee_id	name	department	salary
Filter...	Filter...	Filter...	Filter...
3	Rahul	Finance	45000
4	Priya	IT	50000



employee_id	name	department	salary
Filter...	Filter...	Filter...	Filter...
4	Priya	IT	50000

Task 6: Data Formatting, Sorting & Filtering

Using an Excel dataset: Apply proper data formatting. Format headers and data appropriately. Sort the dataset. Apply filters. Filter records based on specific conditions. Submit screenshots showing the formatted and filtered dataset.

--output---

Employee ID	Name	Age	Department	Salary
1	Aarav	25	IT	₹ 40,000.00
2	Riya	24	HR	₹ 35,000.00
3	Rahul	27	Finance	₹ 45,000.00
4	Priya	26	IT	₹ 50,000.00
5	Karan	28	Marketing	₹ 42,000.00

DATA SORTING ACCORDING TO SALARY

Employee ID	Name	Age	Department	Salary
4	Priya	26	IT	₹ 50,000.00
3	Rahul	27	Finance	₹ 45,000.00
5	Karan	28	Marketing	₹ 42,000.00
1	Aarav	25	IT	₹ 40,000.00
2	Riya	24	HR	₹ 35,000.00

ONLY IT DEPARTMENT FILTER

Employee ID	Name	Age	Department	Salary
4	Priya	26	IT	₹ 50,000.00
1	Aarav	25	IT	₹ 40,000.00

Task 7: Conditional Formatting & Excel Functions

Using an appropriate dataset, demonstrate: Conditional Formatting Highlight values based on conditions. Functions IF() COUNTIF() SUMIF() Use each function at least once and explain its purpose

-----------------------------------explanation---

Conditional Formatting: Automatically highlights cells when they meet a specified condition.

IF(): Checks a condition and returns one value if the condition is true and another value if it is false.

COUNTIF(): Counts the number of cells that meet a specified condition.

SUMIF(): Adds the values that meet a specified condition.

---------------------------------------output--

SALARY >40000

Employee ID	Name	Age	Department	Salary
1	Aarav	25	IT	₹ 40,000.00
2	Riya	24	HR	₹ 35,000.00
3	Rahul	27	Finance	₹ 45,000.00
4	Priya	26	IT	₹ 50,000.00
5	Karan	28	Marketing	₹ 42,000.00

Employee ID	Name	Age	Department	Salary	Salary Status
1	Aarav	25	IT	₹ 40,000.00	Normal Salary
2	Riya	24	HR	₹ 35,000.00	Normal Salary
3	Rahul	27	Finance	₹ 45,000.00	High Salary
4	Priya	26	IT	₹ 50,000.00	High Salary
5	Karan	28	Marketing	₹ 42,000.00	High Salary

Description	Result
Number of IT Employees	2

Description	Result
Total IT Salary	90000

Task 8: VLOOKUP & XLOOKUP

Create or use two related Excel tables and: Use VLOOKUP() to retrieve information. Use XLOOKUP() to retrieve information. Demonstrate the difference in usage where appropriate. Submit screenshots of the results.

-----------------------------------explanation--

VLOOKUP(): Searches for a value in the first column of a table and retrieves the corresponding information from another column.

XLOOKUP(): Searches for a value in one range and returns the corresponding value from another range.

Difference: XLOOKUP is more flexible because the lookup and return ranges are specified separately, while VLOOKUP uses a column number to determine which data to return.

--output--

Employee ID	Name	Age	Department	Salary
1	Aarav	25	IT	₹ 40,000.00
2	Riya	24	HR	₹ 35,000.00
3	Rahul	27	Finance	₹ 45,000.00
4	Priya	26	IT	₹ 50,000.00
5	Karan	28	Marketing	₹ 42,000.00

Employee ID	Employee Name
1	Aarav
3	Rahul
5	Karan

Employee ID	Employee Name
2	Riya
4	Priya

Task 9: Pivot Table & Charts

Using an Excel dataset: Create a Pivot Table. Summarize the data using appropriate fields. Create at least one chart based on the analyzed data. Give a brief explanation of the chart.

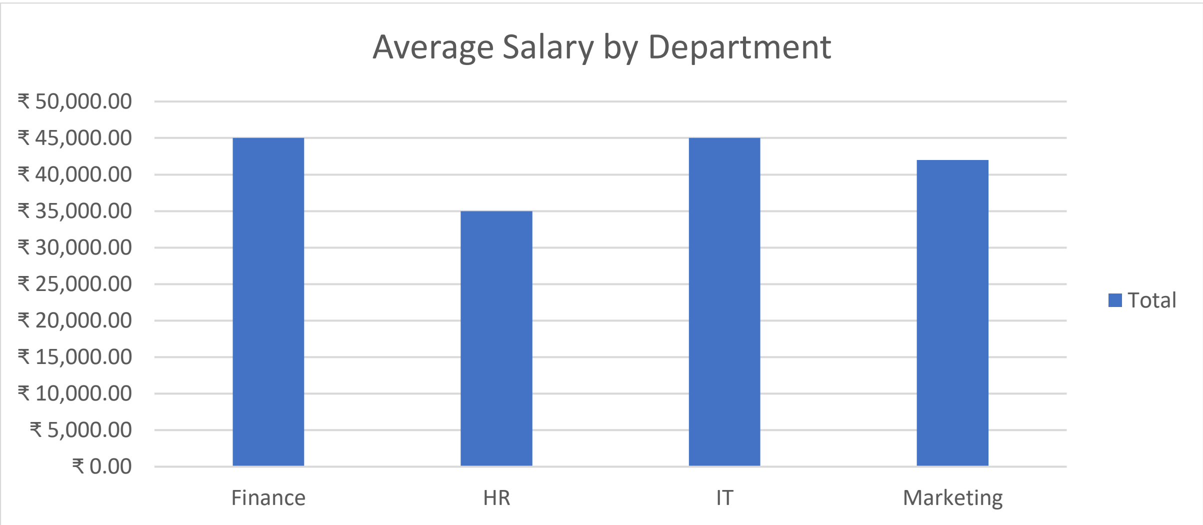
------------------------------------explanation---

A Pivot Table is used to summarize and analyze large amounts of data by grouping information into categories.

The chart provides a visual representation of the summarized data, making it easier to compare values between different departments. In this task, the chart shows the total salary for each department, making it easy to identify which department has the highest and lowest total salary.

--output---

Row Labels	Average of Salary
Finance	₹ 45,000.00
HR	₹ 35,000.00
IT	₹ 45,000.00
Marketing	₹ 42,000.00
Grand Total	**₹ 42,400.00**



❖Code link

https://drive.google.com/drive/folders/1ICXvHjrgZ-rGHfJcpfRFnC3S60nHEAeX?usp=sharing